

國立清華大學
National Tsing Hua University

碩士論文
Master's Thesis

論文標題
Thesis Title

系所別：資訊工程學系

Dept./Grad. Inst.：Department of Computer Science

學號 Student ID.：XXXXXXXXXX

研究生 Author：呂宸漢 Chen-Han Lu

指導教授 Advisor：派大星 Patrick Star

中華民國 一一五 年 七 月
July 2026

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摘要

這裡寫你的摘要...

Abstract

Your abstract goes here ...

誌謝

這裡寫你的誌謝...

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Chapter 1

Introduction

1.1 Motivation

1.2 Previous Works

1.3 Our Contributions

1.4 Thesis Organization

Chapter 2

Preliminaries

Your preliminaries go here ...

Chapter 3

Methodology

Your methodology goes here ...

Chapter 4

Experimental Results

4.1 Experiment Setup

4.2 Results

Chapter 5

Conclusion

Your conclusion goes here ...

Chapter 6

Template and Manual

6.1 This is a section.

6.1.1 This is a subsection.

This is a subsubsection.

6.2 Label

The `\label{...}` in LaTeX is used to label the object. We can use `\ref{...}` to cite the object. This is one way to categorize each label type.

- `\label{ch:...}`: for chapter.
- `\label{sec:...}`: for section, subsection or subsubsection.
- `\label{fig:...}`: for figure.
- `\label{tab:...}`: for table.
- `\label{eq:...}`: for equation.
- `\label{alg:...}`: for algorithm.

6.3 Figure and Table

You can use `\begin{figure} [...]` or `\begin{table} [...]` to determine where to place your figure or table. The available options to fill in `[...]` are listed as follows:

- h: place here.
- t: place on the top of a page.
- b: place on the bottom of a page.
- p: place on a new page.

6.3.1 Figure

You can cite a figure like this: Figure 6.1.



Figure 6.1: Example of a single figure.

You can cite the whole figure using `\ref{fig:...}` (Figure 6.2) or cite the subfigure using `\subref{fig:...}` (Figure 6.2(a)).



(a) Caption of subfigure 1.



(b) Caption of subfigure 2.

Figure 6.2: Example of two subfigures.

6.3.2 Table

You can use the LaTeX table generator (<https://www.tablesgenerator.com/>) to generate a table and paste it here.

Table 6.1 is a table in traditional style.

Table 6.1: Traditional Table Style

Benchmark	Baseline		Ours	
	Score	Runtime (s)	Score	Runtime (s)
testcase1	12	100	11	1
testcase2	18	150	13	1.5

Table 6.2 is a table in booktabs style.

Table 6.2: Booktabs Table Style

Benchmark	Baseline		Ours	
	Score	Runtime (s)	Score	Runtime (s)
testcase1	12	100	11	1
testcase2	18	150	13	1.5

You can use the resize box (`\resizebox{0.9\textwidth}{!}{}{}`) to cover the whole `\begin{tabular} ... \end{tabular}` to adjust the size of a table as Table 6.3.

Table 6.3: Table after Resizing

Benchmark	Baseline		Ours	
	Score	Runtime (s)	Score	Runtime (s)
testcase1	12	100	11	1
testcase2	18	150	13	1.5

If your table is too wide to fit in a page width, you can use a sideways table Table 6.4.

Table 6.4: Sideways Table

Benchmark	Baseline		Ours	
	Score	Runtime (s)	Score	Runtime (s)
testcase1	12	100	11	1
testcase2	18	150	13	1.5

6.4 Equation

Use $\$ \dots \$$ to make a simple variable or mathematical function, such as a , A , $\mathcal{A}$, α , $\sum_{i=0}^n x_i$, ... There are some ways to insert equations:

- `\begin{equation}`: insert a single equation.
- `\begin{gather}`: insert multiple equations. Use `\\` to finish a line and start a new line.
- `\begin{align}`: insert multiple equations and align. Use `\\` to finish a line and start a new line. Use `&` as an anchor to align your equations.

Note that adding an asterisk (*) to any of the environments above (e.g., `align*`) will create an unnumbered version.

The following equation is numbered. We can cite it using `\eqref{eq: \dots}` as Equation (6.1).

$$Cost \leftarrow Cost_{\text{wirelength}} + Cost_{\text{via}} \quad (6.1)$$

These are a group of numbered equations:

$$Cost_{\text{wirelength}} \leftarrow \text{get wirelength cost} \quad (6.2)$$

$$Cost_{\text{via}} \leftarrow \text{get via cost} \quad (6.3)$$

$$Cost \leftarrow Cost_{\text{wirelength}} + Cost_{\text{via}} \quad (6.4)$$

These are a group of unnumbered equations with alignment:

$$Cost_{\text{wirelength}} \leftarrow \text{get wirelength cost}$$

$$Cost_{\text{via}} \leftarrow \text{get via cost}$$

$$Cost \leftarrow Cost_{\text{wirelength}} + Cost_{\text{via}}$$

6.5 Algorithm

Algorithm 1 is an example of writing pseudo code in LaTeX.

Algorithm 1: CalWL: Calculate the total wirelength

Input: A group of nets $\mathcal{N}$ and a group of cells $\mathcal{C}$

Output: Wirelength W

```
1  $W \leftarrow 0$ ;  
  /* Calculate the wirelength for each net */  
2 foreach net  $n \in \mathcal{N}$  do  
3    $(x_1, y_1, x_2, y_2) \leftarrow \text{GetBoundary}(n)$ ;  
   // Wirelength is measured in Manhattan distance.  
4    $w \leftarrow (x_2 - x_1) + (y_2 - y_1)$ ;  
5    $W \leftarrow W + w$ ;  
6 end  
  
  /* Get the boundary of a net */  
7 Function GetBoundary (net  $n$ ) :  
8    $x_1 \leftarrow \infty$ ;  $y_1 \leftarrow \infty$ ;  $x_2 \leftarrow 0$ ;  $y_2 \leftarrow 0$ ;  
9   foreach cell  $c \in \mathcal{C}(n)$  do  
10     $x_1 \leftarrow \min(x_1, x_c)$ ;  $y_1 \leftarrow \min(y_1, y_c)$ ;  
11     $x_2 \leftarrow \max(x_2, x_c)$ ;  $y_2 \leftarrow \max(y_2, y_c)$ ;  
12  end  
13  return  $(x_1, y_1, x_2, y_2)$ ;  
14 end  
15 return  $W$ ;
```

6.6 Itemize

Items with serial numbers:

1. Item 1 ...
2. Item 2 ...
3. Item 3 ...

Unnumbered items:

- Item 1 ...
- Item 2 ...
- Item 3 ...

6.7 Citation

Generate your references in BibTeX format and put them in “reference.bib”. Then, you can use `\cite{...}` to cite your reference, like Chang et al. [1]. It is recommended to use Google Scholar (<https://scholar.google.com.tw/>) to get your references in BibTeX format.

References

- [1] Y.-J. Chang, Y.-T. Lee, J.-R. Gao, P.-C. Wu, and T.-C. Wang, “Nthu-route 2.0: a robust global router for modern designs,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 29, no. 12, pp. 1931–1944, 2010.